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Project:

Smart Parking Management System

Project Overview

The Smart Parking Management System is a database-driven solution designed to improve the efficiency, organization, and management of parking facilities. The system provides a centralized platform for managing users, vehicles, parking zones, parking slots, bookings, payments, parking passes, dynamic pricing policies, and user reviews.

The primary objective of the system is to simplify parking operations for both users and administrators. Users can register their accounts, manage their vehicles, reserve parking slots, purchase parking passes, make payments, and provide feedback regarding parking services. Administrators can oversee parking zones, manage parking slot availability, configure dynamic pricing rules, and monitor overall parking activities.

The system is implemented using a relational database model to ensure efficient data storage, retrieval, and management. Relationships among entities are maintained through primary and foreign keys, which reduce redundancy and improve data consistency. The database is normalized up to Third Normal Form (3NF) to ensure integrity, scalability, and maintainability.

The Smart Parking Management System aims to provide a reliable, scalable, and user-friendly solution that enhances parking management, optimizes parking space utilization, and improves the overall parking experience.

Problem Statement

The rapid increase in the number of vehicles in urban areas has created significant challenges in managing parking facilities efficiently. Traditional parking systems often

depend on manual record keeping, paper-based parking slips, and unorganized monitoring methods. These approaches frequently lead to inefficient utilization of parking spaces, inaccurate record management, payment tracking issues, and difficulties in maintaining parking operations.

Drivers often face problems finding available parking spaces, resulting in unnecessary delays, traffic congestion, fuel consumption, and frustration. Parking administrators also face challenges in managing users, vehicles, bookings, payments, parking passes, and parking slot availability through manual methods.

In addition, the absence of a centralized parking database makes it difficult to maintain accurate records, generate reports, implement flexible pricing policies, and analyze parking trends. Such limitations reduce operational efficiency and negatively affect the overall parking experience.

To address these issues, a Smart Parking Management System is required. The proposed system automates parking management processes by maintaining a structured relational database that stores and manages parking-related information efficiently. The system provides an organized platform for managing users, vehicles, bookings, payments, parking passes, parking reviews, and dynamic pricing policies while ensuring data integrity and consistency.

Proposed Smart Parking Solution

The proposed Smart Parking Management System provides an integrated database solution that automates parking management activities and maintains all parking-related information in a centralized environment.

The system allows users to register their accounts and manage multiple vehicles. Users can reserve available parking slots, purchase parking passes according to their needs, make payments for bookings, and submit reviews regarding parking services.

Administrators are responsible for managing parking zones, parking slots, pass plans, pricing policies, and overall parking operations. Dynamic pricing rules can be configured to support different parking rates based on specific time periods, categories, and day types.

The system maintains relationships among users, vehicles, bookings, payments, parking passes, reviews, and parking zones through a normalized database structure. This ensures efficient data retrieval, accurate record management, and improved operational control.

By automating parking operations and reducing manual intervention, the system improves parking efficiency, enhances user convenience, and supports effective decision-making through organized data management.

Project Scope

The scope of the Smart Parking Management System includes the design and implementation of a relational database for managing parking operations. The system focuses on efficient data storage, retrieval, processing, and management through a normalized database structure.

The project covers the following areas:

User Management

- User registration and account management.
- Secure storage of user information.
- Maintenance of user profiles.

Vehicle Management

- Registration of multiple vehicles per user.
- Classification of vehicles through vehicle types.
- Maintenance of vehicle-related information.

Parking Management

- Creation and management of parking zones.
- Allocation and monitoring of parking slots.
- Maintenance of parking slot availability and status.

Booking Management

- Reservation of parking slots by users.
- Recording of parking duration and booking status.
- Maintenance of booking history.

Payment Management

- Recording and tracking of payment transactions.
- Storage of payment status and transaction references.
- Revenue management and reporting support.

Pass Management

- Creation of parking pass plans.
- Subscription and management of parking passes.
- Tracking pass validity and status.

Dynamic Pricing Management

- Definition of pricing policies for parking zones.
- Support for time-based and category-based pricing.
- Flexible parking charge management.

Review Management

- Submission of ratings and reviews by users.
- Collection of customer feedback.
- Service quality evaluation.

Administrative Management

- Centralized management of parking operations.
- Monitoring of bookings, payments, passes, and pricing policies.
- Maintenance of parking infrastructure records.

The project does not include real-time IoT sensor integration, EV charging management, GPS navigation, mobile notifications, artificial intelligence-based parking prediction, or

third-party payment gateway integration. These features may be considered for future enhancements.

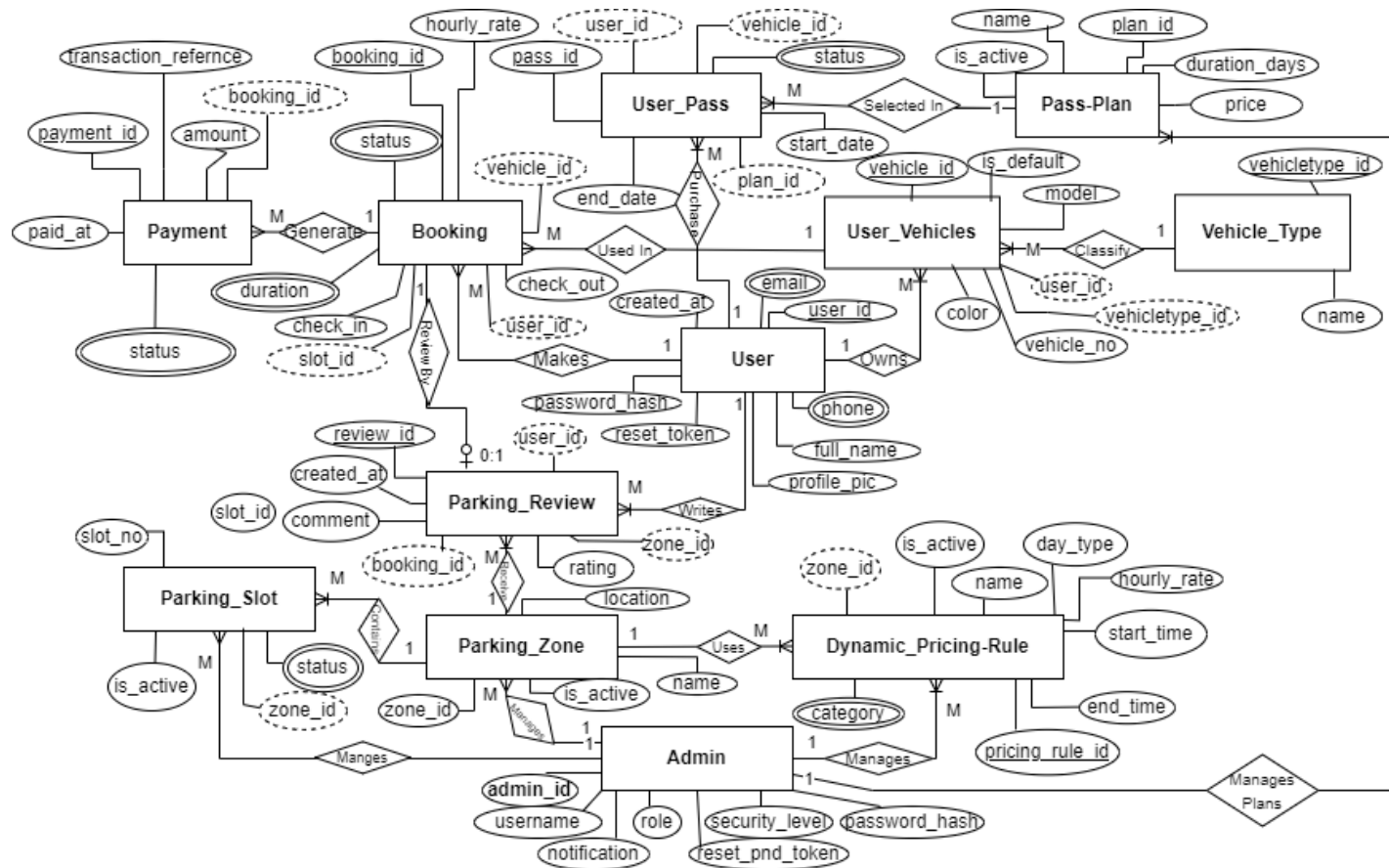
Expected Outcomes

The Smart Parking Management System is expected to provide an efficient and organized platform for managing parking facilities through a structured relational database.

The expected outcomes of the project include:

1. Efficient management of users, vehicles, bookings, payments, parking passes, and parking reviews.
 2. Improved utilization of parking zones and parking slots through organized booking management.
 3. Reduction of data redundancy through normalization up to Third Normal Form (3NF).
 4. Enhanced data integrity through the implementation of primary keys, foreign keys, unique constraints, check constraints, and default values.
 5. Accurate tracking of parking bookings, payments, and parking pass subscriptions.
 6. Flexible parking charge management through dynamic pricing rules.
 7. Efficient data retrieval using SQL queries involving filtering, aggregation, grouping, joins, and subqueries.
 8. Better reporting and decision-making capabilities through structured data management.
 9. Scalability to support future expansion of parking zones, users, vehicles, and parking services.
 10. A maintainable and reliable database system that supports effective parking operations and improves overall user satisfaction.
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ER Diagram (Chen Notation)



Database Normalization

Normalization Goals

Normalization is the process of organizing data within a database to reduce redundancy, eliminate data anomalies, and improve data integrity. The Smart Parking Management System database has been designed and normalized up to Third Normal Form (3NF) to ensure efficient storage, consistency, and maintainability of data.

The database manages users, vehicles, parking zones, parking slots, bookings, payments, parking passes, dynamic pricing rules, and parking reviews. Since these entities are highly interconnected, normalization is essential to avoid duplication of information and maintain structured relationships among entities.

The normalization process ensures that each piece of information is stored only once, updates can be performed without affecting data consistency, and relationships between entities are maintained through foreign keys. As a result, the database becomes easier to maintain, scalable for future expansion, and capable of supporting efficient query processing and reporting.

First Normal Form (1NF)

A relation is in First Normal Form (1NF) if:

- Each attribute contains only atomic (indivisible) values.
- There are no repeating groups or multi-valued attributes.
- Each record is uniquely identified by a primary key.

In the Smart Parking Management System:

- Every table contains a unique primary key such as `user_id`, `vehicle_id`, `booking_id`, `payment_id`, `zone_id`, `slot_id`, and `plan_id`.
- Attributes such as `full_name`, `email`, `phone`, `vehicle_number`, `slot_number`, `transaction_reference`, and `username` contain only single values.
- Multiple vehicles owned by a user are stored as separate records in the **USER_VEHICLES** table instead of storing multiple vehicle numbers in the **USERS** table.
- Multiple bookings made by a user are maintained as separate records in the **BOOKINGS** table.
- Multiple parking passes are stored individually in **USER_PASSES** rather than being grouped within user records.
- Reviews, payments, and pricing rules are stored in separate relations without repeating groups.

Examples:

USERS:

Each user record stores one name, one email, and one phone number.

USER_VEHICLES:

Each vehicle record represents a single vehicle associated with a user.

BOOKINGS:

Each booking record represents one parking reservation.

PAYMENTS:

Each payment record represents one payment transaction.

Since all attributes contain atomic values and repeating groups have been eliminated, all relations satisfy First Normal Form (1NF).

Second Normal Form (2NF)

A relation is in Second Normal Form (2NF) if:

- It is already in First Normal Form (1NF).
- All non-key attributes are fully dependent on the entire primary key.
- There are no partial dependencies.

In the Smart Parking Management System:

All tables use single-column primary keys such as:

- user_id
- admin_id
- vehicle_id
- vehicle_type_id
- zone_id
- slot_id
- booking_id
- payment_id
- plan_id
- pass_id
- review_id
- pricing_rule_id

Because no table contains a composite primary key, partial dependency cannot occur.

Examples:

USERS:

user_id → full_name, email, password_hash, phone

USER_VEHICLES:

vehicle_id → vehicle_number, model, color, is_default

BOOKINGS:

booking_id → check_in_time, check_out_time, duration_hours, hourly_rate,
total_payment, status

PAYMENTS:

payment_id → amount, paid_at, status, transaction_reference

PASS_PLANS:

plan_id → name, duration_days, price, is_active

Each non-key attribute depends completely on its respective primary key and not on part of a key.

Therefore, all relations satisfy Second Normal Form (2NF).

Third Normal Form (3NF)

A relation is in Third Normal Form (3NF) if:

- It is already in Second Normal Form (2NF).
- There are no transitive dependencies.
- Non-key attributes depend only on the primary key and not on other non-key attributes.

The Smart Parking Management System satisfies Third Normal Form by separating related information into independent entities and linking them through foreign keys.

Examples:

Vehicle Information:

Vehicle type information is stored separately in VEHICLE_TYPES and referenced through USER_VEHICLES using vehicle_type_id. This prevents repeated storage of vehicle type names throughout the database.

Parking Infrastructure:

Parking zone information is maintained in PARKING_ZONES while parking slot information is maintained in PARKING_SLOTS. Bookings reference parking slots using foreign keys rather than duplicating zone and slot information.

Booking Management:

BOOKINGS stores only booking-specific information and references users, vehicles, and parking slots through foreign keys. User names, vehicle details, and parking zone details are not duplicated within booking records.

Payment Management:

Payment details are stored separately in PAYMENTS and linked to bookings through booking_id. This avoids storing payment information repeatedly within booking records.

Pass Management:

PASS_PLANS stores plan details such as duration and price, while USER_PASSES references pass plans through plan_id. This eliminates duplication of plan information.

Review Management:

PARKING_REVIEWS references users, bookings, and parking zones through foreign keys rather than storing duplicate user or booking information.

Dynamic Pricing:

Pricing policies are maintained separately in DYNAMIC_PRICING_RULES and associated with parking zones through zone_id. This avoids repeating pricing information across multiple tables.

By separating data into specialized entities and maintaining relationships through foreign keys, redundancy is minimized, update anomalies are eliminated, and data consistency is maintained.

Therefore, the Smart Parking Management System database satisfies Third Normal Form (3NF).

Functional Dependencies

Table	Functional Dependency
USERS	user_id → full_name, email, password_hash, phone, created_at
ADMINS	admin_id → username, password_hash, role, notification_email, security_level
VEHICLE_TYPES	vehicle_type_id → name
USER_VEHICLES	vehicle_id → user_id, vehicle_type_id, vehicle_number, model, color, is_default
PARKING_ZONES	zone_id → name, location, is_active
PARKING_SLOTS	slot_id → zone_id, slot_number, status, is_active
BOOKINGS	booking_id → user_id, vehicle_id, slot_id, check_in_time, check_out_time, duration_hours, hourly_rate, total_payment, status
PAYMENTS	payment_id → booking_id, amount, status, paid_at, transaction_reference
PASS_PLANS	plan_id → name, duration_days, price, is_active
USER_PASSES	pass_id → user_id, vehicle_id, plan_id, start_date, end_date, status

Table	Functional Dependency
PARKING_REVIEWS	review_id → user_id, zone_id, booking_id, rating, comment, created_at
DYNAMIC_PRICING_RULES	pricing_rule_id → zone_id, name, day_type, start_time, end_time, hourly_rate, is_active, category

Relational Schema

Schema Definition

The Smart Parking Management System database consists of twelve interconnected entities designed to manage parking operations efficiently.

USERS stores customer account information, while **ADMINS** maintains administrator credentials and access levels. **USER_VEHICLES** stores vehicle information associated with users and **VEHICLE_TYPES** categorizes different vehicle classes.

PARKING_ZONES and **PARKING_SLOTS** represent the parking infrastructure. **BOOKINGS** records parking reservations made by users and **PAYMENTS** maintains financial transactions associated with bookings.

PASS_PLANS defines subscription packages, while **USER_PASSES** records pass ownership and validity periods. **PARKING_REVIEWS** stores customer feedback and ratings regarding parking services.

DYNAMIC_PRICING_RULES manages flexible pricing policies based on parking zones, time periods, and categories.

Relationships among these entities are maintained through primary and foreign key constraints, ensuring referential integrity, reducing redundancy, and supporting efficient retrieval of information across the entire system.

USERS(user_id (PK), full_name NOT NULL, email UNIQUE NOT NULL, password_hash NOT NULL, phone UNIQUE, created_at DEFAULT CURRENT_TIMESTAMP, profile_pic_url, reset_pwd_token)

ADMINS(admin_id (PK), username UNIQUE NOT NULL, password_hash NOT NULL, role NOT NULL, notification_email UNIQUE, security_level DEFAULT 1, reset_pwd_token)

VEHICLE_TYPES(vehicle_type_id (PK), name UNIQUE NOT NULL)

USER_VEHICLES(vehicle_id (PK), user_id (FK) NOT NULL, vehicle_type_id (FK) NOT NULL, vehicle_number UNIQUE NOT NULL, model NOT NULL, color, is_default DEFAULT FALSE)

PARKING_ZONES(zone_id (PK), name UNIQUE NOT NULL, location NOT NULL, is_active DEFAULT TRUE)

PARKING_SLOTS(slot_id (PK), zone_id (FK) NOT NULL, slot_number NOT NULL, status DEFAULT 'Available', is_active DEFAULT TRUE)

BOOKINGS(booking_id (PK), user_id (FK) NOT NULL, vehicle_id (FK) NOT NULL, slot_id (FK) NOT NULL, check_in_time NOT NULL, check_out_time, duration_hours CHECK (duration_hours >= 0), hourly_rate CHECK (hourly_rate >= 0), total_payment CHECK (total_payment >= 0), status DEFAULT 'Active')

PAYMENTS(payment_id (PK), booking_id (FK) UNIQUE NOT NULL, amount CHECK (amount >= 0), status DEFAULT 'Pending', paid_at, transaction_reference UNIQUE)

PASS_PLANS(plan_id (PK), name UNIQUE NOT NULL, duration_days CHECK (duration_days > 0), price CHECK (price >= 0), is_active DEFAULT TRUE)

USER_PASSES(pass_id (PK), user_id (FK) NOT NULL, vehicle_id (FK) NOT NULL, plan_id (FK) NOT NULL, start_date NOT NULL, end_date NOT NULL, status DEFAULT 'Active')

PARKING_REVIEWS(review_id (PK), user_id (FK) NOT NULL, zone_id (FK) NOT NULL, booking_id (FK) UNIQUE NOT NULL, rating CHECK (rating BETWEEN 1 AND 5), comment, created_at DEFAULT CURRENT_TIMESTAMP)

DYNAMIC_PRICING_RULES(pricing_rule_id (PK), zone_id (FK) NOT NULL, name NOT NULL, day_type NOT NULL, start_time NOT NULL, end_time NOT NULL, hourly_rate CHECK (hourly_rate >= 0), is_active DEFAULT TRUE, category NOT NULL)
